# SafeLens

An Evidence-Aware Multilingual AI Safety Assistant for Community Incident Intelligence

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## 1. Introduction

People share safety information through social media, messaging groups, phone calls and word of mouth. In Karachi, these reports are often written in English, Urdu, Roman Urdu or a mix of all three. They are frequently incomplete, repeated, exaggerated or contradictory, which makes it hard to see what is actually happening and which incidents need attention first.

SafeLens is a web-based safety assistant that uses Artificial Intelligence to turn these unstructured reports into structured incident records. It groups related reports into a single evolving incident thread and shows how well each claim is supported.

## 2. Background

Several products already offer parts of this functionality, including Police 15 Karachi, SafeKarachi, SafetyMapper and WatchTower. SafeLens does not claim to be the first safety app. It focuses on a different problem: improving how uncertain, multilingual reports are understood and combined.

## 3. Problem Statement

Most crowd-reporting systems display each submission as a separate alert or map marker. This can:

- create duplicate incidents for the same event,
- amplify unverified claims,
- hide disagreement between witnesses,
- make an area look unsafe based on limited or biased data.

Users and responders need to know not only what was reported, but also what is supported, what is uncertain, and what is contradicted.

## 4. Proposed Solution

SafeLens accepts a short written report in English, Urdu, Roman Urdu or mixed language, together with a location. The AI converts the report into structured data, compares it with existing reports, and either creates a new incident thread or adds the report to an existing one. Each thread displays a timeline and claim-level evidence labels. The result appears on an interactive map and in a simulated responder view, along with cautious, situation-specific safety guidance.

SafeLens is designed to complement official emergency services, not replace them.

## 5. Key Innovation: Evidence-Aware Incident Threading

Instead of treating every submission as a new event, SafeLens links reports using meaning, time, location and incident type. It then labels each claim in the thread as:

- **Supported by multiple reports:** several users independently describe the same detail.
- **Unconfirmed:** only one report mentions it.
- **Contradictory:** reports disagree.

The system never declares a report to be proven true. It makes uncertainty visible.

## 6. Objectives

1. Accept safety reports in English, Urdu, Roman Urdu and mixed language.
2. Use AI to extract incident type, location, time, severity, a short summary and a recommended action.
3. Detect duplicate and related reports and group them into incident threads.
4. Track escalation within a thread, for example from suspicious activity to possible robbery.
5. Label claims by level of support instead of declaring reports true or false.
6. Store reports and incidents in a database.
7. Present incidents through an interactive map, an incident dashboard and a responder view.
8. Highlight high-severity incidents with basic alerts.

## 7. Project Scope

### Included

- Web-based report form (text and location)
- Multilingual AI extraction and summarization
- Related-report matching and incident threads
- Escalation timeline
- Claim labels and contradiction detection
- Interactive map and dashboard
- Responder view showing status, reports, timeline and evidence
- Basic alerts for high-severity incidents

### Outside the initial scope

- Voice, image and video input (listed as future work)
- Direct emergency service dispatch
- Continuous real-time GPS tracking
- A production-level mobile application
- Predictive policing or surveillance

## 8. System Workflow

Citizen Report → Language Detection → Structured AI Extraction → Claim Confidence and Contradiction Analysis → Related-Report Matching → Create or Update Incident Thread → Database → Map and Responder View → Cautious Safety Guidance

## 9. Technology Stack

| Component | Tool |
|---|---|
| Language | Python |
| Interface and dashboard | Streamlit |
| AI analysis | Groq API (LLM) |
| Database | SQLite |
| Map | Folium and OpenStreetMap (streamlit-folium) |
| Geocoding | geopy with Nominatim |
| Data handling and charts | pandas |
| API key management | python-dotenv |
| Code and documentation | GitHub |
| Interface design | Figma or Canva |

## 10. Methodology

The project follows an iterative approach, building and testing one feature at a time:

1. Database design and AI extraction
2. Report form and storage
3. Related-report matching and incident threads
4. Claim labels and contradiction detection
5. Map and dashboard
6. Responder view and interface polish
7. Testing and documentation

## 11. Prototype Scope and Testing

The prototype will be tested with approximately 25 simulated reports covering:

- minor, moderate and major incidents,
- English, Urdu, Roman Urdu and mixed-language submissions,
- duplicated and related reports,
- corroborating and contradictory statements,
- escalating incidents,
- missing or unclear location details.

**Minimum demonstration:** submit a report, view the AI structured output, see it grouped with a related report, see an escalation update, view the incident on the map, and open the responder view.

## 12. Target Users

- Community members reporting incidents
- Administrators and responders reviewing reports
- Safety organizations that need structured incident information
- Students and researchers studying AI-based safety applications

## 13. Significance

The project combines Artificial Intelligence, natural language processing, database management and map visualization. It addresses a gap in existing tools by handling multilingual, uncertain and conflicting reports in a transparent way, which is especially relevant to the language mix used in Pakistan.

## 14. Ethical Principles

- User-supplied reports are evidence inputs, not automatic proof.
- The system does not use facial recognition, identify people as criminals, make accusations, predict crime or dispatch police automatically.
- Official emergency contacts will come only from verified sources.
- Alerts will clearly state when information is unverified.

## 15. Limitations

- The prototype uses simulated data.
- AI may misunderstand incomplete, unclear or heavily mixed-language reports.
- Location accuracy depends on what users provide.
- Analysis depends on internet and API availability.
- Alerts are basic system alerts, not full emergency notifications.
- The prototype is for academic demonstration and is not a replacement for official emergency services.

## 16. Expected Outcome

A working end-to-end prototype:

Report → AI → Structured Incident → Database → Related Reports → Escalation → Map → Responder View

Its main value is transparency. It shows what is reported, what is supported, what is disputed and what still needs verification.

## 17. Future Work

Voice reporting, image and video evidence analysis, real-time notifications, a mobile application, and integration with verified official data sources.

## 18. Conclusion

SafeLens turns scattered and uncertain citizen reports into one explainable incident story. By combining multilingual AI analysis, evidence-aware threading and clear uncertainty labels, it offers a practical and responsible foundation for future community safety tools.

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### Supervisor Pitch

*“SafeLens is not just another crime map. It is a multilingual, evidence-aware system that turns scattered and uncertain citizen reports into one explainable incident story, showing what is reported, what is supported, what is disputed, and what still needs verification.”*